

Clique-Size Dominance Threshold Distribution

Enumerating Labeled Graphs by Clique Number

Revised edition with corrigendum and verification supplement

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Abstract

We enumerate labeled simple graphs on n vertices by their clique number, giving the triangle $d(n, k)$ of graphs with clique number exactly k . The triangle is recorded in the OEIS as A395684; its bounded-multiplicity extensions and cumulative distribution are A395691–A395695. It provides an exact finite distribution of clique sizes and, under the chosen clique-dominance convention $K > \omega(G)$, of the threshold $K_{\text{dom}} = \omega(G) + 1$. We give verified values for $n \leq 8$, exact near-diagonal formulas, the weighting formula for labeled p -multigraphs, and reproducible computational and formal verification material. This revision retains the enumerative results and all independently reproduced sequence terms. A dated corrigendum clarifies the precision of the random-graph approximation, the expansion discussion, related OEIS comparisons, an overlap expression, and the distinction between a chosen dominance threshold and a necessary minimum for lossless decoding.

1 Introduction

The Clique-Cluster (CC) transformation replaces each input vertex with a coding clique and represents an edge of multiplicity q by q same-index edges between its coding cliques [1]. For a simple input graph G , choosing a coding size $K > \omega(G)$ ensures that coding cliques are larger than transversal cliques. The least integer satisfying this imposed dominance condition is

$$K_{\text{dom}}(G) = \omega(G) + 1.$$

This is a prescribed coding rule, rather than a claim that no smaller lossless encoding or different decoder is possible.

The natural enumerative question is how clique numbers are distributed over all $2^{\binom{n}{2}}$ labeled graphs on n vertices. The present triangle answers that question exactly. Its unlabeled counterpart is OEIS A263341, using graph complementation. The labeled triangle and its extensions were accepted as A395684 and A395691–A395695 in May 2026 [2]. Their definitions and counts are independent of the CC interpretation and of random-graph asymptotics.

The resulting distributions are relevant to finite graph enumeration, K_{k+1} -free graph counting, and analysis of the chosen coding rule. Section 11 records the clarifications to the May 2026 edition. These clarifications do not alter the finite enumerative contribution.

2 Definitions

Let $\mathcal{G}_n$ denote the simple graphs on the fixed labeled vertex set $[n] = \{1, \dots, n\}$, with $n \geq 1$.

Definition 2.1 (Clique number). For $G = ([n], E)$, define

$$\omega(G) = \max\{|S| : S \subseteq [n], \binom{S}{2} \subseteq E\}.$$

Here $\binom{S}{2}$ is the set of unordered pairs of distinct vertices of S .

Definition 2.2 (Dominance threshold). The clique-dominance threshold is $\tau(G) = \omega(G) + 1$. Thus $\tau(G)$ is the least integer K satisfying $K > \omega(G)$. For simple graphs we also write $K_{\text{dom}}(G) = \tau(G)$.

Definition 2.3 (Exact distribution). For $1 \leq k \leq n$, define

$$d(n, k) = |\{G \in \mathcal{G}_n : \omega(G) = k\}|.$$

Set $d(n, k) = 0$ outside this range. These are the rows of A395684.

Definition 2.4 (Cumulative distribution). Define

$$D(n, k) = \sum_{j=1}^k d(n, j), \quad D(n, 0) = 0.$$

Equivalently, $D(n, k)$ counts graphs with clique number at most k . These are the rows of A395695.

3 Main Results

3.1 Structural properties

Theorem 3.1 (Boundary values). For $n \geq 1$, $d(n, 1) = d(n, n) = 1$.

Proof. A graph has clique number one exactly when it has no edges, so the empty graph is the unique member of the first bucket. A graph on n vertices has clique number n exactly when it is complete. $\square$

Theorem 3.2 (Row sum). For $n \geq 1$,

$$\sum_{k=1}^n d(n, k) = 2^{\binom{n}{2}}.$$

Proof. Each of the $\binom{n}{2}$ possible edges is present or absent independently in specifying a labeled graph. The clique-number buckets partition this finite set. $\square$

3.2 Cumulative counts and inclusion–exclusion

Theorem 3.3 (Difference formula). Let $c(n, k)$ count K_{k+1} -free labeled graphs. For $1 \leq k \leq n$,

$$D(n, k) = c(n, k), \quad d(n, k) = c(n, k) - c(n, k-1),$$

with $c(n, 0) = 0$ and $c(n, n) = 2^{\binom{n}{2}}$.

Proof. A graph is K_{k+1} -free exactly when its clique number is at most k . Subtracting graphs with clique number at most $k-1$ leaves those with clique number exactly k . $\square$

For completeness, ordinary inclusion–exclusion gives

$$c(n, k) = \sum_{\mathcal{F} \subseteq \binom{[n]}{k+1}} (-1)^{|\mathcal{F}|} 2^{\binom{n}{2} - |\bigcup_{S \in \mathcal{F}} \binom{S}{2}|}. \quad (1)$$

Indeed, the intersection of clique events indexed by $\mathcal{F}$ forces exactly the edges in the displayed union; every other edge is free. The empty family contributes $2^{\binom{n}{2}}$.

3.3 Near-diagonal formulas

Theorem 3.4 (Near-diagonal closed forms). *For $n \geq 2$,*

$$\begin{aligned} d(n, n-1) &= n2^{n-1} - \binom{n}{2} - n, \\ c(n, n-2) &= 2\binom{n}{2} - n2^{n-1} + \binom{n}{2} + n - 1. \end{aligned}$$

Proof. For each vertex v_i , let A_i be the event that $[n] \setminus \{v_i\}$ is a clique. Then $|A_i| = 2^{n-1}$, and $|A_i \cap A_j| = 2$ for $i \neq j$: only the edge $\{v_i, v_j\}$ remains free. An intersection of three or more distinct events forces the complete graph and has size one. Thus

$$\left| \bigcup_i A_i \right| = n2^{n-1} - 2\binom{n}{2} + \sum_{j=3}^n (-1)^{j+1} \binom{n}{j} = n2^{n-1} - \binom{n}{2} - n + 1.$$

Removing the complete graph proves the first formula. Subtracting $d(n, n-1) + d(n, n)$ from the total gives the second. The empty higher intersection sum at $n = 2$ gives the same identities. $\square$

Remark 3.1 (Overlap bookkeeping). General intersections depend on the entire overlap pattern. For two $(k+1)$ -subsets intersecting in t vertices, the forced-edge count is

$$2\binom{k+1}{2} - \binom{t}{2}.$$

More generally it is the cardinality of the edge union in (1), or equivalently

$$\sum_{\emptyset \neq I \subseteq \{1, \dots, r\}} (-1)^{|I|+1} \binom{|\bigcap_{i \in I} S_i|}{2}.$$

This corrects the informal overlap expression in the original Remark 3.1; the tabulations and near-diagonal proof do not use that expression.

3.4 Extension to bounded-multiplicity graphs

Definition 3.1 (p -multigraph). A labeled p -multigraph assigns a multiplicity $\mu : \binom{[n]}{2} \rightarrow \{0, 1, \dots, p\}$. Its clique number is that of the underlying simple graph with edge set $\{e : \mu(e) > 0\}$.

Theorem 3.5 (Weighted distribution). *For $p \geq 1$, the number $d_p(n, k)$ of these multigraphs with clique number k is*

$$d_p(n, k) = \sum_{\substack{G \in \mathcal{G}_n \\ \omega(G) = k}} p^{e(G)}.$$

Proof. Each present edge of a fixed support graph G has p possible positive multiplicities. Nonedges have multiplicity zero. Hence G has exactly $p^{e(G)}$ preimages. Summing over supports proves the formula. $\square$

Corollary 3.1 (Multigraph row sum). $\sum_{k=1}^n d_p(n, k) = (p+1)\binom{n}{2}.$

Proof. Each unordered vertex pair has $p+1$ multiplicity choices, including zero. Clique-number buckets partition those assignments. $\square$

4 Computational Results

The original tables below are retained. Additional verification in October 2026 independently reproduced every term of all six sequences through $n = 8$. Two distinct clique-number methods enumerated all $2^{28} = 268,435,456$ labeled eight-vertex simple graphs and agreed with the stored weighted rows for $p = 1, \dots, 5$.

4.1 Labeled simple graphs by clique number

$n \backslash k$	1	2	3	4	5	6	7	8
1	1							
2	1	1						
3	1	6	1					
4	1	40	22	1				
5	1	387	570	65	1			
6	1	5788	21837	4970	171	1		
7	1	133500	1353096	576247	33887	420	1	
8	1	4682269	142458868	110940707	10151470	201152	988	1

The totals are $2^{\binom{n}{2}}$, and the penultimate entries agree with Theorem 3.4; for example $d(8, 7) = 988$.

4.2 Cumulative distribution

$n \backslash k$	1	2	3	4	5	6	7	8
1	1							
2	1	2						
3	1	7	8					
4	1	41	63	64				
5	1	388	958	1023	1024			
6	1	5789	27626	32596	32767	32768		
7	1	133501	1486597	2062844	2096731	2097151	2097152	
8	1	4682270	147141138	258081845	268233315	268434467	268435455	268435456

The column $D(n, 2)$ counts labeled triangle-free graphs and agrees with A213434. Each final entry is the total $2^{\binom{n}{2}}$.

4.3 Labeled 2-multigraphs by clique number

$n \backslash k$	1	2	3	4	5	6	7
1	1						
2	1	2					
3	1	18	8				
4	1	248	416	64			
5	1	6264	36080	15680	1024		
6	1	294986	6434528	6536000	1050624	32768	
7	1	25126170	2555681112	5913294912	1833638912	130514944	2097152

These totals are $3^{\binom{n}{2}}$. Complete verification data for all $p = 1, \dots, 5$ through $n = 8$ are in the repository.

4.4 Dominance-threshold distribution

For simple graphs, the number receiving $K_{\text{dom}} = k + 1$ under the specified dominance rule is exactly $d(n, k)$: it is the same table with shifted column labels. For a multigraph with actual maximum multiplicity P , the safe rule in the cited construction is $K = \max(P, \omega(G)) + 1$; clique number alone does not specify its entire parameter choice.

4.5 Verification scope

The supplied clique oracle was checked against explicit vertex-subset testing for all 33,867 graphs with $1 \leq n \leq 6$. Separate exhaustive runs checked all 2,097,152 seven-vertex graphs and all eight-vertex graphs. The resulting (ω, e) histograms reproduce all five weighted distributions and their exact totals. Cumulative rows are checked by termwise prefix sums. The later published rows, through $n = 12$ for $p = 1$ and $n = 11$ for $p = 2, \dots, 5$, additionally pass endpoint, total and near-diagonal checks. Their larger interior entries were not exhaustively recomputed in the October verification. These limits are recorded explicitly with the reproducibility material.

5 OEIS Cross-References

- A395684: the labeled clique-number triangle $d(n, k)$.
- A395691–A395694: $d_p(n, k)$ for $p = 2, 3, 4, 5$ respectively.
- A395695: the cumulative triangle $D(n, k)$.
- A263341: the unlabeled analogue, equivalently by independence number under complementation. At $n = 4, k = 2$ its entry is 6, compared with the labeled count $d(4, 2) = 40$.
- A058843: labeled graphs equipped with a proper partition into k nonempty color classes, with permutations of color names identified. It is not a distribution by chromatic number. Its $n = 4$ row is 1, 80, 192, 64.
- A006125: the total $2^{\binom{n}{2}}$, the row sums of $d(n, k)$.
- A213434: labeled triangle-free graphs, the column $D(n, 2)$.
- A000088: unlabeled simple graph totals, the row sums of A263341.

Clique number and chromatic number are different invariants: the five-cycle has clique number two and chromatic number three. For four vertices, however, every triangle-free graph is bipartite, so both exact-two counts equal 40. The earlier value 31 was not a valid chromatic-number comparison.

6 Algorithmic Implementation

The independent verification records, for each edge count e , the number $H_n(k, e)$ of graphs with clique number k . It then evaluates

$$d_p(n, k) = \sum_{e=0}^{\binom{n}{2}} H_n(k, e) p^e.$$

One oracle tests candidate vertex subsets directly for all required edges. A second uses the induced-subgraph recurrence

$$\omega(G[S]) = \max(\omega(G[S \setminus \{v\}]), 1 + \omega(G[(S \setminus \{v\}) \cap N(v)])).$$

This follows by separating cliques that omit v from those that contain it. The eight-vertex run enumerates all seven-vertex graphs and all possible neighborhoods of the last vertex. The repository includes portable scripts, saved expected terms, per-term comparisons, result files and exact-arithmetic checks. Assertions compare clique buckets themselves; totals are additional checks rather than substitutes for classification verification.

7 Formal Verification

The revised development uses Lean 4.19.0 and mathlib v4.19.0, with pinned dependencies and compilation treating warnings as errors. Its graph distribution is defined using genuine simple graphs on $\text{Fin}(n)$ and their clique number. The proved results include clique-number bounds, the empty and complete graph characterizations, boundary values, out-of-range zeros, cumulative prefix sums and differences, and an explicit equivalence between graphs and sets of unordered edges. This gives the graph total $2^{\binom{n}{2}}$ and both unweighted and weighted row sums. The count $p^{e(G)}$ of positive multiplicity assignments to a fixed support is also proved.

The construction modules define bounded symmetric loopless multiplicities and the same-index CC wiring. They prove the mixed-clique exclusion, identification of coding clusters as exactly the K -cliques under $K > \omega(G)$, and multiplicity recovery when the coding size has sufficient capacity. A separate development proves that, for $K > p$, the closed neighborhoods of minimum cardinality recover exactly all coding clusters, without a clique-number assumption. This formalizes the recovery argument in Section 11. The strict-dominance threshold $\omega + 1$ is proved as the least integer satisfying that inequality, without asserting a necessary minimum for every decoder.

The auxiliary arithmetic module proves parameter bounds, selected census properties and exact CC edge-count identities. Its census counts parameter pairs; no graph-isomorphism classification is asserted. The repository’s formalization inventory and compiler records state the qualifications of each theorem. The revised modules contain no admitted proofs or additional axiom declarations. The general inclusion–exclusion and near-diagonal arguments are proved mathematically in this paper but are not mechanized here. The random-graph asymptotics are cited classical results. Computational reproduction of the finite rows is reported separately from the formal structural identities.

8 Applications and Interpretation

The primary application is the exact distribution of a standard graph invariant across finite labeled graphs. The cumulative values give counts of K_{k+1} -free graphs, and the weighted extensions give the analogous finite census with bounded edge multiplicities. These are concrete enumerative objects independent of any encoding.

For the chosen clique-dominance coding rule, $d(n, k)$ also gives the number of simple graphs assigned each coding size $k + 1$. It can therefore describe that convention’s allocation costs. It does not claim optimality over all graph encodings or decoding rules. Connections with extremal and Ramsey questions require their own additional arguments; a finite clique-number distribution alone does not determine a Fraïssé limit or establish performance on an empirical network population.

9 Asymptotic Analysis

Proposition 9.1 (Clique number of a uniform random graph). *For $G \sim G(n, 1/2)$,*

$$\Pr(\omega(G) = k) = \frac{d(n, k)}{2^{\binom{n}{2}}}.$$

The first-order approximation is

$$\frac{\omega(G)}{2 \log_2 n} \xrightarrow{\Pr} 1.$$

More precisely,

$$\omega(G) = 2 \log_2 n - 2 \log_2 \log_2 n + O(1) \quad \text{with high probability.}$$

There is an integer sequence $k_n = 2 \log_2 n - 2 \log_2 \log_2 n + O(1)$ such that $\Pr(\omega(G) \in \{k_n, k_n + 1\}) \rightarrow 1$.

Proof. Every labeled graph is equiprobable, proving the exact identity. The asymptotic statements are the classical clique-number results of Bollobas and Erdos, including the finer concentration statement [3, 4]. $\square$

Proposition 9.2 (Expansion under the dominance rule). *For the chosen simple-graph rule $K_{\text{dom}} = \omega(G) + 1$,*

$$\mathbb{E}[K_{\text{dom}}] = 2 \log_2 n - 2 \log_2 \log_2 n + O(1).$$

Its expected output vertex count is

$$n [2 \log_2 n - 2 \log_2 \log_2 n + O(1)].$$

Thus the expected vertex count is $\Theta(n \log n)$, a logarithmic expansion factor relative to the input's n vertices.

Proof. Use the classical expectation estimate for the clique number [3, 4] and the identities $K_{\text{dom}} = \omega + 1$ and $|V(f_{CC}(G))| = nK_{\text{dom}}$ for this rule. $\square$

This is a statement about the specified uniform model and coding convention. It does not by itself establish runtime efficiency, optimal encoding size, or typical behavior of real-world graphs.

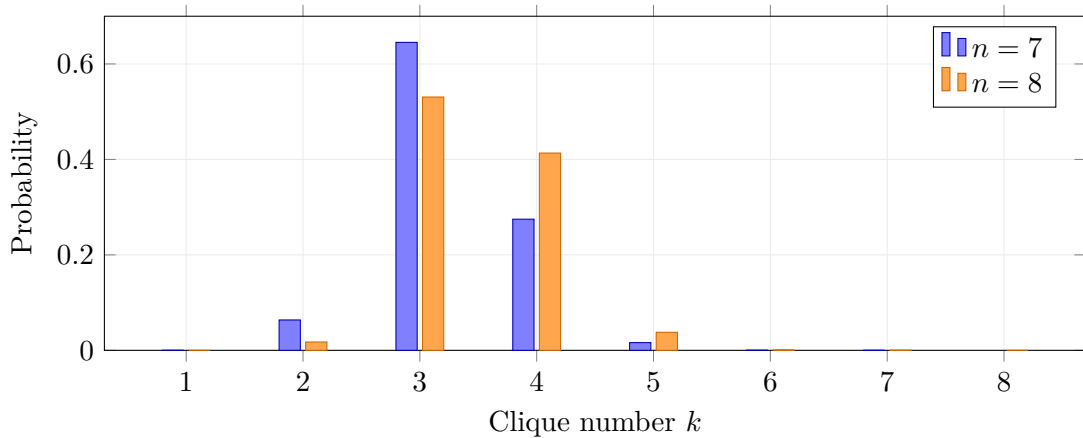


Figure 1: Exact clique-number probabilities for labeled graphs on seven and eight vertices, obtained by normalizing the retained enumeration rows. Both rows have mode $k = 3$. At $n = 8$ the mean is 3.473740656. These finite distributions illustrate the enumerative result; the finer asymptotic location is stated in Proposition 9.1.

10 Conclusion

The triangles $d(n, k)$, $d_p(n, k)$ and $D(n, k)$ give exact finite counts by clique number, bounded multiplicity and cumulative clique bound. Their original tabulated values remain unchanged and are supported by additional independent termwise reproduction. The clique-dominance threshold $k + 1$ remains a well-defined interpretation under the chosen coding convention. Separating that convention from necessary minimum encoding size, and separating first-order asymptotics from precise concentration, makes the scope of the result explicit without changing its enumerative content.

11 Corrigendum to the May 2026 Edition

This section records clarifications identified in an author-initiated review, supported by additional independent verification. The finite enumeration is the paper’s main result. Its definitions, retained tables, near-diagonal closed forms and multigraph weighting formula are unchanged. None of the corrections below changes any independently reproduced term of A395684 or A395691–A395695.

11.1 Precision of the original Proposition 9.1

The expression $2 \log_2 n$ is a valid first-order approximation to the clique number of $G(n, 1/2)$. In the original Proposition 9.1, that approximation was used more precisely than justified by saying the probability concentrates on its floor or ceiling. The actual concentration location includes the unbounded second-order term $-2 \log_2 \log_2 n$, as now stated in Proposition 9.1. The exact identity $\Pr(\omega = k) = d(n, k)/2^{\binom{n}{2}}$ is unchanged. The approximation was not used to compute the finite tables.

11.2 Expansion caveat and finite-data caption

For the specified dominance rule, the first-order output estimate $n(2 \log_2 n + 1)$ describes the leading scale. The more precise expectation is given in Proposition 9.2. The phrase “only marginally larger” is replaced by the explicit description “a logarithmic expansion factor.” Statements about a uniform random-graph model do not establish performance on empirical network distributions. Figure 1 now describes the actual finite rows rather than using the first-order asymptotic expression as their concentration center.

11.3 Related sequence identifications

A058843 counts properly colored labeled graphs, identifying permutations of color names; it does not count graphs by their chromatic number. Its $n = 4, k = 2$ entry is 80. The original chromatic-number comparison using 31 is replaced by the correct description in Section 5. The corresponding unlabeled $n = 4, k = 2$ entry of A263341 is 6, rather than the neighboring entry 3. These were comparison descriptions, not inputs to the enumeration algorithms. The labeled value $d(4, 2) = 40$ is unchanged.

11.4 Dominance convention and lossless recovery

The value $k + 1$ is the least integer satisfying the imposed condition $K > k$. It should not be read as a necessary minimum for every lossless decoder. Indeed, the cited same-index wiring admits the following elementary recovery method when $K > P$, with P the maximum multiplicity.

Every vertex of a coding clique has its $K - 1$ internal neighbors. The unused index K has no external neighbors, so the minimum degree of the image is exactly $K - 1$. Each vertex of this degree has closed neighborhood equal to its coding clique. Taking the distinct such neighborhoods recovers the entire cluster partition; counting the edges between each pair recovers its multiplicity. Thus $K = P + 1$ suffices for this decoder without a bound on the input clique number. For simple graphs, $K = 2$ suffices.

The triangle therefore measures clique number and the assigned threshold under the dominance convention, not an optimal cost over all decoding methods. This corrects the necessity/minimum interpretation while preserving the enumeration and the sufficient largest-clique identification argument. It also avoids the earlier tension between a chosen coding rule and general recoverability.

11.5 Overlap expression and formal status

Remark 3.1 in the original edition contained an incorrect overlap expression. The correct two-set subtraction is $\binom{t}{2}$ when the vertex intersection has size t ; Remark 3.1 now gives the edge-union formulation. That expression was not used to generate the retained tables or prove the near-diagonal formulas. The updated formalization material states its exact proved scope, separately from cited asymptotics and finite computational checks. Historical sources and the original editions remain available in the repository archive.

11.6 Reproducibility and preservation of the result

The supplementary verification compares individual entries, not only row totals. It independently reproduces all six distributions through eight vertices, including weighted and cumulative values. The corrected comparisons, asymptotic precision, overlap remark and decoding interpretation do not enter those computations. The accepted finite enumerative contribution and its use as a labeled-graph clique-number census are therefore preserved.

References

- [1] A. Zarzuelo Urdiales, *Canonical Embedding of Universal p -Graphs, Commensurable Weighted Graphs, and Line Structures into the Rado Graph via Clique-Size Dominance Transformation*, December 2025.
- [2] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>; A395684, A395691–A395695, A263341, A058843, A006125 and A213434.
- [3] B. Bollobas and P. Erdos, “Cliques in random graphs,” *Mathematical Proceedings of the Cambridge Philosophical Society* **80** (1976), 419–427. https://www.renyi.hu/~p_erdos/1976-05.pdf.
- [4] B. Bollobas, *Random Graphs*, 2nd ed., Cambridge University Press, 2001.
- [5] A. Zarzuelo Urdiales, *Clique-OEIS: source, formalization and reproducibility material*, <https://github.com/alejandrozu/Clique-OEIS>.
- [6] A. Zarzuelo Urdiales, *Clique-Size Dominance Threshold Distribution*, May 2026, original edition, <https://doi.org/10.5281/zenodo.20004703>.